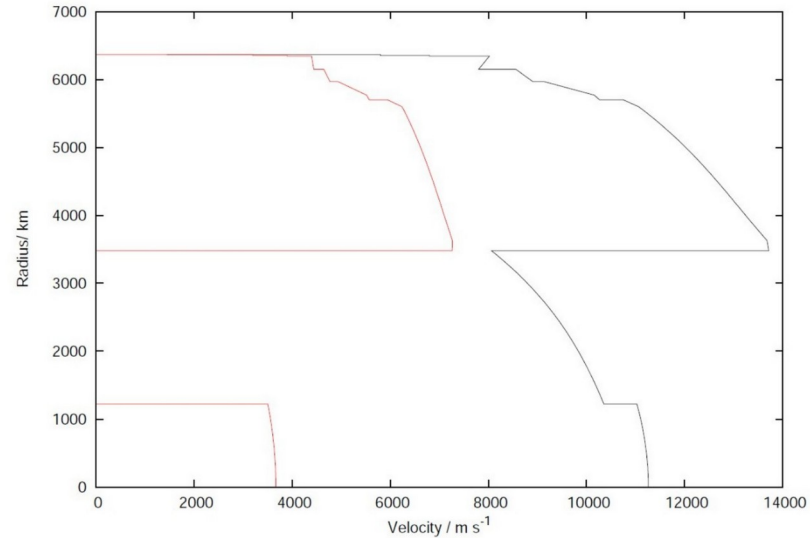
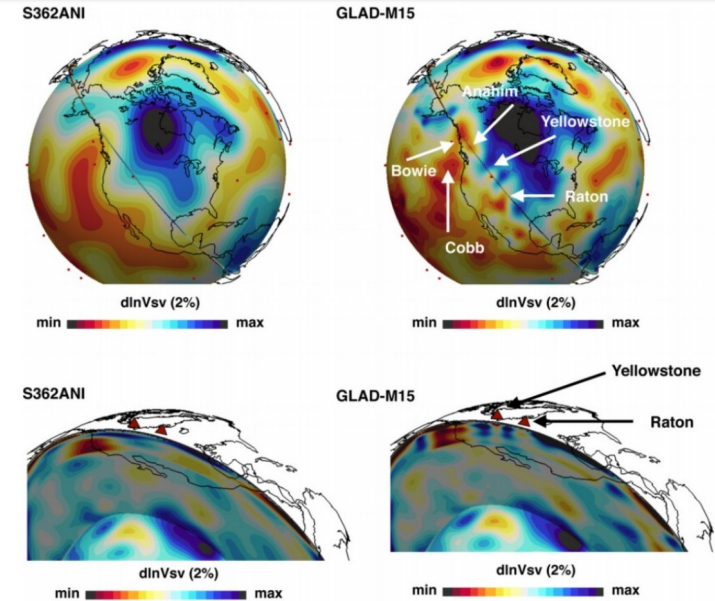


Self gravitation and rotation

From earlier lectures



Preliminary Reference Earth Model (PREM) of Dziewonski & Anderson (1981)



Bozdag *et al.* 2016 – first global waveform inverse model

Newtonian gravity

Poisson's equation

Gravitational potential satisfies:

$$\nabla^2 \phi = 4\pi G \rho.$$

Associated gravitational acceleration:

$$g_i = -\frac{\partial \phi}{\partial x_i}$$

Equation subject to continuity conditions:

$$[\phi]_{-}^{+} = 0, \quad [g_i \hat{n}_i]_{-}^{+} = 0$$

across any jumps in density, while the potential also tends to zero at infinity

Poisson's equation

Solution to the problem can be expressed as a familiar integral:

$$\phi(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3\mathbf{y}',$$

and differentiating, the gravitational acceleration can be similarly written:

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3\mathbf{y}'.$$

Transformation to the reference body

Using the motion of the body, we can define a **referential gravitational potential**:

$$\zeta(\mathbf{x}, t) = \phi[\varphi(\mathbf{x}, t), t],$$

Using conservation of mass, we can transform the integral solution to obtain:

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d\mathbf{x}',$$

Similarly for the referential gravitational acceleration:

$$\gamma_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]}{\{[\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)][\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)]\}^{\frac{3}{2}}} d\mathbf{x}',$$

Gravitational binding energy

This is the potential energy of a body associated with gravitational forces

Can be calculated by assuming the body is assembled bit-by-bit, summing up the work required to move each particles in from infinity relative the to current field

Calculation identical to that in 1B Electrodynamics for the energy of an electrostatic field:

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \varrho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3\mathbf{y}.,$$

Full details in the notes, but you won't be asked to recall the calculation

Can express referentially:

$$\mathcal{V}_g(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) \zeta(\mathbf{x}, t) d\mathbf{x}.$$

Equations of motion

Hamilton's principle

Potential energy is now the sum of elastic and gravitational contributions

Modified Lagrangian density:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t),$$

Statement of Hamilton's principle is unchanged

Derivation of Euler-Lagrange equations is harder because we have a **non-local** term in the Lagrangian – this reflects the **action at a distance** property of Newtonian gravity

First variation of the action

Functional derivatives still defined through linear term in Taylor expansion

The result can be expressed in the form

$$\langle D\mathcal{S}(\varphi)|\delta\varphi\rangle = \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d\mathbf{x} dt.$$

The “derivative” terms here are, by definition, just the quantities multiplying the first-order perturbations to the fields

In the case of local Lagrangians, these are just normal partial derivatives

It is only the variation with respect to the motion that needs additional work

Euler-Lagrange equations

Follow the standard integrations by parts, we obtain:

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta \mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta \mathcal{L}}{\delta F_{ij}} \right) = 0,$$

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0$$

As previously, we have:

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}$$

In example sheet 4 you will show that:

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i$$

Euler-Lagrange equations

The equations of motion can be written in the form:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0,$$

As expected, we have just added a gravitational body force to the balance of linear momentum

This force is expressed in terms of the motion through an integral over the whole body

These are now **integro partial differential equations**

Co-rotating reference frames

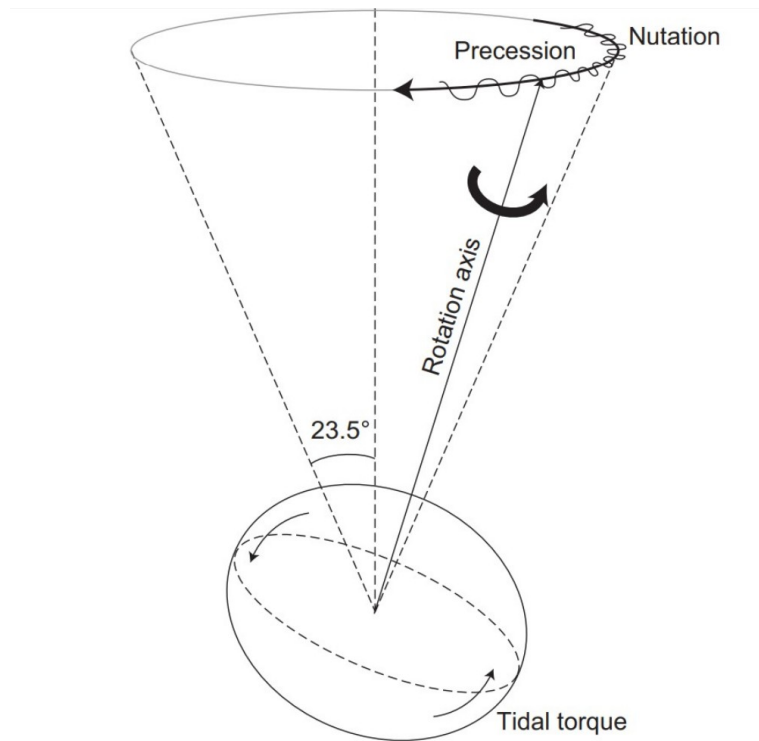
We have described the motion relative to an inertial frame defined by the “fixed stars”

For applications to the Earth, it is more natural to use a rotating frame

The Earth’s rotation is not steady, varying due to both external and internal processes

Within seismology, it is common to use an steady and approximately co-rotating reference frame

This will introduce the usual fictitious forces into the equations of motion



Decomposition of the motion

The motion relative to inertial space can be written:

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t)\bar{\varphi}_j(\mathbf{x}, t),$$

Translation vector represents motion of the origin of the reference frame

Rotation matrix represents rotation of the reference frame

On the right hand side we see the **relative motion** as seen from the non-inertial frame

Because of material frame indifference, the choice of frame has no effect on the potential energy associated with the motion

Relative velocity

Differentiating the motion in time, we find:

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kj} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right)$$

Introduction of the angular velocity as seen from the non-inertial frame:

$$R_{kj} \frac{dR_{kl}}{dt} = \epsilon_{jkl} \Omega_k$$

The velocity can then be expressed in the form:

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} (\bar{v}_j + \epsilon_{jkl} \Omega_k \bar{\varphi}_l).$$

Relative acceleration and choice of frame

Differentiating again, we can write the acceleration:

$$\frac{\partial v_i}{\partial t} = \frac{d^2 \Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl} \Omega_k \bar{v}_l + \epsilon_{jkl} \frac{d\Omega_k}{dt} \bar{\varphi}_l + \epsilon_{jkl} \epsilon_{lmn} \Omega_k \Omega_m \bar{\varphi}_n \right)$$

At this point, we define an approximately co-rotating frame by setting:

$$\frac{d\Phi}{dt} = 0, \quad \frac{d\Omega}{dt} = 0,$$

This approach can be generalised to allow for interactions between different planets (e.g., Earth-Moon system)

Stress and gravity in the non-inertial frame

Differentiating the motion spatially, we find:

$$F_{ij} = R_{ik} \bar{F}_{kj}$$

From the definition of the first Piola-Kirchhoff stress and frame indifference, we find:

$$T_{ij} = R_{ik} \bar{T}_{kj}$$

Similarly, for the referential gravitational acceleration:

$$\gamma_i = R_{ij} \bar{\gamma}_i$$

New equations of motion

Putting things together and cancelling common rotation factors:

$$\rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk}\Omega_j\bar{v}_k + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l\bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho\bar{\gamma}_i = 0.$$

All terms defined relative to the rotating reference frame

Identical to the inertial case except for the addition of **Coriolis and centrifugal forces**

From here on we will always work in the rotating frame and drop the overbars

Linearisation

Same procedure as ever for linearisation on application of a small stress glut

At zeroth-order we have the equilibrium equations:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 = 0,$$

Stress-free equilibrium solutions are not now possible

At first order we have the linearised equations for the displacement vector:

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho \gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

Perturbation to referential gravitational acceleration given as an integral of the displacement vector – see handout for full expression

Scaling analysis

Order of magnitude estimates

Derivatives can be readily estimated in terms of typical time and length scales

$$\begin{aligned}\rho \frac{\partial^2 u_i}{\partial t^2} &\sim \frac{\bar{\rho} U}{T^2}, \\ \rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} &\sim \frac{\bar{\rho} \Omega U}{T}; \\ \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m &\sim 2 \bar{\rho} \Omega^2 U, \\ \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) &\sim \frac{\bar{\mu} U}{L^2},\end{aligned}$$

Estimating the magnitude of the gravitational acceleration is a bit more work due to integral definition (see handout):

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U$$

Relative importance of terms

Ratio of gravitational to elastic forces:

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2},$$

This shows that gravity is negligible at sufficient small wavelengths and hence at correspondingly high frequencies

Taking L to be the radius of the Earth:

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3.$$

At the longest wavelengths gravity is just as important as elasticity within the equations of motion – this fact means that we can learn about density directly

Relative importance of terms

Ratio of rotational to inertial forces:

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho}\Omega U}{T}}{\frac{\bar{\rho}U}{T^2}} = \Omega T$$

If time-scale is much less than a day, rotation will be negligible

Longest period seismic free oscillations have periods on one hour – rotation is not negligible, but is a relatively small effect

For longer period deformation such as that due to tidal forces, rotation becomes a dominant feature within the dynamics

